Will Unexpectedness Help Recommendations and When: Evidence from A Large-Scale Online Controlled Experiment

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Abstract

Personalized recommender systems have achieved great business success by matching targeted users with their interested items. However, the classical recommendation paradigm focused primarily on similarity-based measures without delivering variety of recommended items to the targeted consumers, therefore suffering from the problem of user boredom and filter bubbles.

In this paper, we present a Deep-Learning based unexpected recommendation method and conduct an online controlled experiment testing performance of this method in a large-scale online video platform. The results show that adoption of our unexpected recommendations increases the click-through rates on videos by 3%, video viewing by 8% and the watching time of the videos by 25% vis-à-vis the recommendation methods currently deployed at Alibaba-Youku. Furthermore, we demonstrate that adding moderate levels of unexpectedness to the recommender system would lead to the best performance, compared to adding either low or high levels of unexpectedness. In addition, we show that the treatment effects of adopting unexpected recommendations are heterogeneous among different groups of consumers, whereas some consumers would witness a significant positive impact, while others may even endure a negative impact. The presented recommendation method is currently in the process of being deployed at Alibaba.

Introduction

Recommender systems have been popularized by Amazon and Netflix in 2000's and have been subsequently widely adopted by many firms across various industries over the last several years due to numerous economic benefits that the industry derived from deploying personalized recommendations (Gomez-Uribe and Hunt 2015; Adomavicius & Tuzhilin 2005). In particular, they are shown to be capable of influencing consumer choices (Senecal & Nantel 2004), generating the lift in sales (De et al. 2010), enhancing consumer trust (Panniello et al. 2016) and providing other economic benefits to consumers (Gomez-Uribe and Hunt 2015).

Although useful in many cases, recommender systems often lead to the "filter bubble" problem (Pariser 2011; Nguyen et al. 2014; Resnick et al. 2013), limiting personalized recommendations to those items with which the users are already familiar and feel comfortable, thus isolating them in their own interest bubbles. This filter bubble phenomenon is undesirable because it can significantly reduce sales diversity (Fleder & Hosanagar 2009) and can lead to

the problem of user boredom (Kapoor et al. 2015), which significantly deteriorates user satisfaction with recommender systems.

To address these concerns and to improve user satisfaction with recommendations, researchers have proposed to incorporate the concept of unexpectedness in recommender systems (Murakami et al. 2007; Ge et al. 2010), which aims at providing novel and useful recommendations at the same time. One particular way to provide unexpected recommendations is through multi-objective optimization between usefulness objective and unexpectedness objective (Adamopoulos & Tuzhilin 2011, Adamopoulos & Tuzhilin 2014). Although there have been a few alternative approaches to defining unexpectedness proposed over the last several years that are reviewed in Section 2, few of them have been successfully implemented in the industry, and whether unexpectedness can improve recommendations in practical business settings still remains an unanswered question.

In this paper, we present a Deep Learning based unexpected recommendation method that defines unexpectedness in the latent space of item embeddings and test its performance in an online controlled experiment conducted on over half a million customers for the period of one week for the video recommendation application at Alibaba-Youku, one of the largest video and streaming providers in China serving more than 500 million users (Alibaba 2020). We demonstrate that the adoption of our unexpected recommendation method leads to a significant lift of 3% in the overall video consumptions across the platform. Therefore, we show that incorporating our model into the Alibaba-Youku recommendation platform indeed encourages users to expand their viewing interests through enhanced interactions with the recommender system. This in turn produces strong economic effect for Alibaba, leading the company to move our unexpected recommendation method into production.

In addition, we show that it is not always better to provide more unexpected recommendations to the targeted users and that effect is the strongest when recommendations are not too unexpected, i.e., when they still take the users outside of their comfort zone without pushing them into the realm of watching irrelevant videos.

Third, we investigate the heterogenous treatment of unexpected recommendations on different types of consumers according to their demographics. In particular, we demonstrate that unexpected recommendations are useful not for all the customers of Alibaba-Youku, but only for certain segments of its population, such as young customers and premium users. This implies that unexpected recommendations should not be provided in the "shotgun" approach to everybody but only to the targeted segments of customers and only in a "measured" way without too much surprise to them, i.e., without risking to make these unexpected recommendations irrelevant.

In this paper we make the following contributions. First, we use our unexpected recommendation method for providing video recommendations at Alibaba-Youku and demonstrate in a controlled experiment that it has a strong economic impact on the performance of the recommender system currently used at Alibaba-Youku. This is the first practical demonstration of the power of unexpected recommendations since the prior literature on unexpectedness only provided various definitions of the concept and presented a few unexpected algorithms that have never been properly tested in the industrial settings. Second, we demonstrate that the unexpected recommendations should not be provided in a mindless "shotgun" approach to everybody, but only to certain targeted segments of customers and in a

"measured" way without taking them too far outside of the comfort zones of their filter bubbles.

Literature Review

Researchers have pointed out that accuracy is not the only important objective of recommendations (McNee et al. 2006), while other beyond-accuracy metrics should also be taken into account, including unexpectedness, serendipity, novelty, diversity, coverage and so on (Ge et al. 2010; Kaminskas et al. 2016). Although these metrics are related to each other, there are still significant conceptual differences among them (Adamopoulos & Tuzhilin 2014). Since we focus on the unexpected recommendations in this paper, we will review in this section mainly the unexpected recommendations literature that tends to recommend those items that differ significantly from user expectations (Adamopoulos & Tuzhilin 2014) , rather than recommending those items that match user interests, as is typically done by classical collaborative filtering methods.

In particular, several papers have proposed ways to incorporate unexpectedness directly into the recommendation process, which can be classified into rule-based approaches (Taramigkou et al. 2013; Lee & Lee 2015), model-based approaches (Wasilewski & Hurley 2019; Kawamae 2010) and hybrid approaches (Adamopoulos & Tuzhilin 2014; Iaquinta et al. 2008). They provide multiple important benefits vis-à-vis the classical recommendation methods, such as overcoming the over-specialization problem, broadening user interests and increasing user satisfaction (Zhang et al. 2012; Adamopoulos & Tuzhilin 2014). There are also numerous marketing literature discussing the benefits of variety-seeking (Zeithammer & Thomadsen 2013; Givon 1984; McAlister & Pessemier 1982; Kahn et al. 1986).

Most of the unexpectedness measures are defined in the feature space of the items (Kotkov et al. 2016; Adamopoulos & Tuzhilin 2014). In contrast, in this paper we model unexpectedness in the latent space of item embeddings, as explained in the next section.

Modeling Unexpectedness

In (Adamopoulos & Tuzhilin 2011, Adamopoulos & Tuzhilin 2014), unexpectedness of a recommended item for a user was measured as its deviation from the items previously consumed by that user in the *feature space*. Then the overall utility of item i to user u is defined as the weighted average between predicted user preferences and the degree of unexpectedness, i.e., $Utility_{u,i} = (1-\alpha) * r_{u,i} + \alpha * Unexp_{u,i}$, where $r_{u,i}$ is the predicted rating of user u of item i, $Unexp_{u,i}$ is the degree of unexpectedness of item i for user u, as previously defined, and α is the "unexpectedness factor" controlling for the extent of unexpectedness in provided recommendations. Although useful, the drawback of this approach lies in the challenges associated with computing the distance between two items in the feature space needed to determine $Unexp_{u,i}$. This is the case because item's individual features can be very heterogeneous, and it is hard to combine distances between very different individual attributes, such as genres of movies and movie's gross revenues, into one common distance function needed to compute $Unexp_{u,i}$.

To address this problem, we propose to map the items into the *latent* space using the AutoEncoder (AE) approach (Diederik & Welling 2014; Hinton & Salakhutdinov 2006) and compute distances between items in the latent space using the Euclidean distance, rather than doing these computations in the feature space. In particular, the autoencoder approach learns

two separate neural networks: the encoder network, which maps explicit features of an object into *latent embeddings*, and the decoder network, which reconstructs the feature vectors from these latent embeddings. Furthermore, we follow a popular approach and represent both the encoder and the decoder as the Multi-Layer Perceptrons (MLP) (Rumelhart et al. 1985) . Specifically, we denote the m-dimensional explicit feature vector of each user as $u = \{u_1, u_2, \dots, u_m\}$, and the n-dimensional explicit feature vector of each item as $i = \{i_1, i_2, \dots, i_n\}$. Then the autoencoder network is optimized through minimizing the following reconstruction loss: $L_{AE} = MLP_{decoder}(MLP_{encoder}(u, i))$. The user and the item embeddings are subsequently obtained as $W_u = MLP_{encoder}(u)$ and $W_i = MLP_{encoder}(i)$ respectively. We subsequently model unexpectedness of user u and item i as the average Euclidean distance between candidate item i, represented as the latent item embedding, and the set of all the previously consumed items u_{past} by user u in the latent space, i.e., $Unexp_{u,i} = \frac{1}{||u_{past}||}\sum_{k \in u_{past}} d(W_i, W_{i_k})$, where W_i and W_{ik} are the latent embedding vector representations of corresponding items. We visualize unexpectedness modeling based on the embedding approach in Figure 1, where the green and the red points represent embeddings W_i and W_{ik} of the previously consumed items respectively. Once we define unexpectedness of item i for user u, we integrate it into the recommendation process through the aforementioned hybrid utility function.

By doing so, we capture hidden and complex interactions between users and items by utilizing powerful capabilities of latent DL-based modeling. This helps us to model user unexpectedness more precisely than it is done in the original feature space, as shown in the literature (He et al. 2017; Zhang et al. 2019; Li & Tuzhilin 2019) and allows us to provide unexpected *and* useful recommendations easier than in the alternative approaches, thus improving user online experiences.

To measure the effectiveness of the proposed approach, we implemented the unexpected recommender system described in this section at Alibaba-Youku, and conducted an online controlled experiment there that we describe in the rest of the paper.

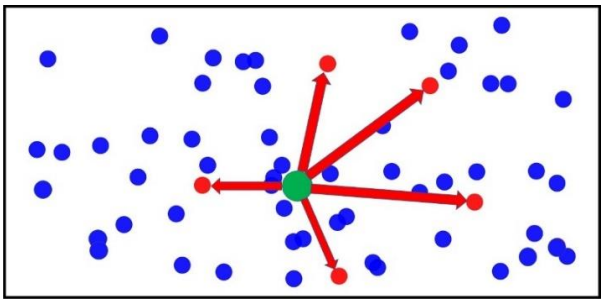

Figure 1: Visualization of the latent modeling of unexpectedness in the 2-dimension.

Description of the Controlled Experiment at Alibaba-Youku

Identification Strategy

As one of the largest online video services in China, Alibaba-Youku has more than 500 million monthly active users, with 800 million daily video views (Alibaba 2020). The objective of the conducted controlled experiment is to identify the Average Treatment Effects (ATEs) of adopting the proposed unexpected recommendation method vis-a-vis the current production

system in terms of user video consumptions.

Note that users have little control over what videos they can watch, while the video exposure is solely determined by the platform through personalized recommendation services. This motivates the choice of Alibaba-Youku for testing our unexpected recommendation method, as it enables us to assess the ATEs directly through the OLS regression by random treatment assignment. During the recommendation process, users are unaware of their assignments, therefore removing any persistent linear, additive consumer-specific effects that may introduce endogeneity due to self-selection.

Heterogenous Treatment Effect

The treatment effect of adopting unexpected recommender system for video consumptions may differ across users, as we are interested in the moderating effect of user demographics, including gender, age groups, residence and their premium status. This interaction effect is theoretically indeterminate, since no existing literature has discussed the user heterogeneity towards unexpected recommendations.

We are also interested in the moderating effect of the degree of unexpectedness towards the user perception of recommended videos. To test for this effect, we incorporate the degree of unexpectedness as moderators of the treatment effect and measure the heterogenous effect by adding the interaction term into the OLS regression.

Methodology

In the controlled experiment, the current production system in Alibaba-Youku is compared to the unexpected recommender system described in Section 3. This is achieved by randomly splitting the users into the control group served by the current production system and the treatment group that receives recommendations from the proposed unexpected recommender system. The user splitting ensures that the demographics and characteristic of users in both the control and the treatment groups are balanced, as validated through the Wilcoxon rank-sum test in Table 1.

The online controlled experiment was performed at Alibaba-Youku serving video recommendations for 26,959,538 user-video interactions over the period of one week from 07/29/2019 to 08/04/2019. The duration of one entire week was agreed upon with the management of Alibaba-Youku, who felt that it would be sufficient to demonstrate the necessary effects of our experiment, and that fluctuations of user behaviors across different weeks were typically insignificant.

Using the server's application protocol interface (API), we record the gender, age, province and city of residence of the 516,201 users in our experimental pool. We also collect information about the operating system of the mobile phone, the premium status of the user (whether the user subscribed to the premium service or not) and the number of days that the user was active during the last month. For each video impression from the 1,589,273 candidate videos, we also record its meta-information, including the number of views, comments and the time since it has been released on Alibaba-Youku.

Variable	Treatment Group	Control Group	p-value
gender	0.005 (0.990)	0.003 (0.990)	0.7040
age group	4.119 (2.100)	4.076 (2.070)	0.1787

province	8.262 (8.930)	8.054 (8.850)	0.3898
city	2.038 (1.550)	2.028 (1.530)	0.8823
operating system	1.172 (0.410)	1.173 (0.400)	0.8809
is_vip	0.361 (0.480)	0.365 (0.480)	0.4492
active_days	20.668 (8.520)	20.404 (8.450)	0.8045

Table 1: User feature-level summary statistics for the control and treatment group.

Business Performance Metrics

Based on the Alibaba guidelines, for each user–item pair (u, i) that is targeted by our unexpected recommender system, we examine the following three business metrics to assess the recommendation performance in the online controlled experiment: (a) Click-Through-Rate (CTR_{ij}) measuring the percentage of user clicks on the recommended video, which equals 1 if user i clicked on video j and 0 otherwise; (b) Video View (VV_{ij}) measuring the amount of videos finished by each user, which equals 1 if user i clicked and *finished* watching video j and 0 otherwise; and (c) Time Spent (TS_{ij}) measuring the amount of time that user i spent on video j. If the user did not click on the video, TS_{ij} would be 0. Note that Alibaba-Youku uses additional business performance metrics on their platform besides these three; but the company requested us to focus on the CTR, VV and TS metrics in this experiment.

Consumption Lift from Adopting Unexpected Recommendations

We start with the regression analysis that directly compares video watching behavior between users in the treatment group and the control group. Specifically, we use the OLS regression to specify the binary outcome variables CTR_{ijt} and VV_{ijt} and the continuous outcome variable TS_{ijt}, as shown below:

$$CTR_{ijt} = \alpha_0 + \alpha_1 Treatment_i + \overrightarrow{\alpha_2}\overrightarrow{X_j} + \overrightarrow{\alpha_3}\overrightarrow{Y_i} + D_t + \varepsilon_{ijt} \quad (1)$$

$$VV_{ijt} = \alpha_0 + \alpha_1 Treatment_i + \overrightarrow{\alpha_2}\overrightarrow{X_j} + \overrightarrow{\alpha_3}\overrightarrow{Y_i} + D_t + \varepsilon_{ijt} \quad (2)$$

$$TS_{ijt} = \alpha_0 + \alpha_1 Treatment_i + \overrightarrow{\alpha_2}\overrightarrow{X_j} + \overrightarrow{\alpha_3}\overrightarrow{Y_i} + D_t + \varepsilon_{ijt} \quad (3)$$

where $Treatment_i$ is a binary variable indicating whether the user i is in the treatment group, $\overrightarrow{X_j}$ represents the explicit user feature vectors, $\overrightarrow{Y_i}$ represents the explicit item feature vectors and D_t represents the time fixed effects including dates and hours. Our results all hold consistent if we cluster standard errors at the user level or at the item level. The results also hold consistent if we use the discrete choice models alternatively, including the logit model and the probit model.

Results

Average Treatment Effect

Table 3 presents the results for the regression models presented above by equations (1) – (3).

The consistent positive and significant coefficients across all the three models indicate that providing unexpected recommendations significantly increases video consumption compared with the control condition, as reflected through the three business metrics. In particular, by interpreting the coefficients in the three models, we conclude that users receiving unexpected recommendations would increase probability of clicking on the recommended videos by 2.74%, probability of finishing watching the video by 1.96%, and the average watching time by 0.7 minute ($p<0.01$ in all cases). Note that the average length of the video is only 3 minutes. Therefore, the unexpected recommender system encourages the user to watch 25% more of the recommended videos on average. Having over 500 million monthly active users watching 800 million videos on a daily basis, our results show very substantial potential increase in business revenues for the company, motivating Alibaba-Youku to move our unexpectedness algorithm into production.

	CTR(1)	VV(2)	TS(3)
Treatment	0.0274*** (0.001)	0.0196*** (0.001)	0.7073*** (0.0937)
R-Squared	0.0045	0.0050	0.0045
Observations	26959538	26959538	26959538

*Table 3: Average Treatment Effect of Unexpected Recommendations. The table shows a regression with robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01*

Effect of Unexpectedness

To assess further how different levels of unexpectedness affect business performance metrics during the recommendation process, we present the study to identify the average treatment effects (ATEs) of the proposed model locally with regard to different unexpectedness levels. More specifically, we aggregate the data records into 27 bins according to the unexpectedness value computed using the method introduced in Section 3, namely [0, 0.05], [0.05, 0.10], [0.10, 0.15], …, [1.30, 1.35]. We select 1.35 as the maximum threshold since there are only very few recommendations in our records whose unexpectedness values are greater than 1.35. We conduct the local regression of three business performance metrics in each bin by adding the additional interaction term $Treatment_i \times Unexp - Level_{ij}$ to the equation (1) and identify the coefficients accordingly. The results are shown in Figure 2(a-c). Note that our findings remain robust when we use a slightly different width to categorize the data records into bins, e.g. 0.4 or 0.6. Specifically, we have the following important observations:

First, providing medium-level unexpected recommendations (unexpectedness values ranging between 0.3 and 0.8) to the users would lead to a significant increase (of 14.04%) in the probability of clicking on the video and significant increase in probability of finishing watching the video (13.41%). This result further supports our claim that unexpected recommendations matter. Furthermore, note that, these performance improvements for moderate unexpected recommendations are significantly bigger than the general improvements across all the recommendations that we presented in the previous section and in Table 3. Moreover, this result is novel and important, suggesting that users are most interested in those unexpected recommendations located just outside of their *"comfort zone,"* i.e., not too close to what they have seen before (i.e., not another episode of Harry Potter or another James Bond movie), but still close to their interests and prior consumptions. In other words, this result shows that the *users do want to break out of their "filter bubble" of prior recommendations*

and consumption.

In contrast, if we provide high-level unexpected video recommendations, the proposed model could *backfire*: we witness a significant *decrease* of click-through rates, video views and time spent watching videos by the users in this case. This indicates that those highly unexpected videos would become *irrelevant* to the users by deviating too far from users' expectations and their "comfort zones." As a result, the users would not see the connections between the recommended videos and their previous watching histories, and they would rather prefer to stay in their comfort zones.

Finally and surprisingly, if we provide low-level unexpected recommendations to the users, this will also significantly deteriorate the three business performance metrics, as is shown in Figure 2. This finding is non-trivial, as previous studies focus heavily on identifying more precise similarity-matching between users and items. Our results, instead, show that traditional similarity-based recommendations are not really meeting users' satisfaction, as those recommendations tend to be redundant and boring, thus in turn hurting the performance of recommendation agents. Instead, targeted users are looking for those "unexpected" recommendations that could broaden their interests and provide them with fresh new contents.

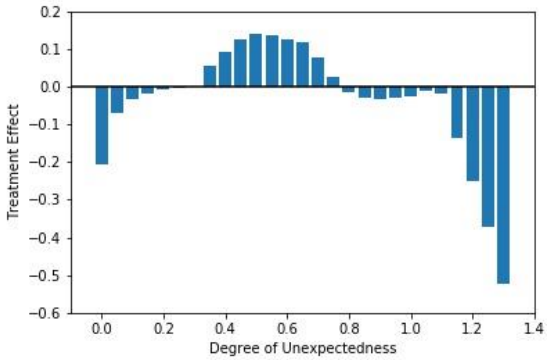

Figure 2 (a) Treatment effect towards CTR over different degree of unexpectedness

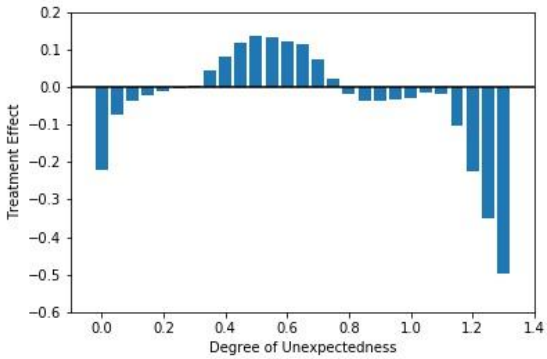

Figure 2 (b) Treatment effect towards VV over different degree of unexpectedness

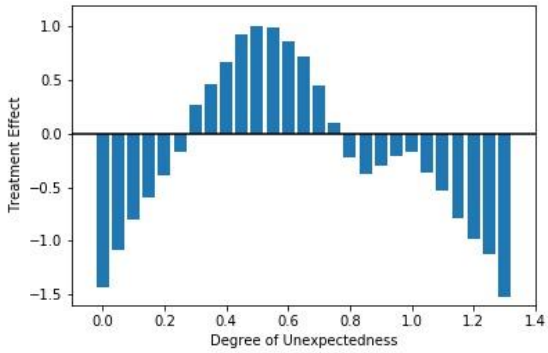

Figure 2 (c) Treatment effect towards TS over different degree of unexpectedness

User Heterogeneity in Treatment Effects

In this section, we report exploratory analysis about the heterogeneous treatment effects of our promotion program across various user demographics. Based on the existing literature and our data availability, we identify a number of characteristics of consumers that may influence the effectiveness of unexpected recommendations on consumer video consumptions. The moderators include (1) **Age group** (2) **Premium Status** (3) **Gender** (4) **Residence (5) Operating System**. We add interactions as variables of interest between the aforementioned moderators and our unexpected recommendation treatment in equations (1) – (3) to the three business metrics.

Table 5(a) shows that the young people (age under 30) are more willing to explore unexpected video contents on the platform, and thus receive a significant positive effect from the treatment of unexpected recommender system much greater than that reported in Table 3. However, for the elder generation (above 60), unexpectedness would actually backfire, as we witness a significant negative effect, indicating that they would rather prefer to enjoy their familiar contents and stay in their "comfort zones". Meanwhile, middle-aged users would also significantly benefit from the unexpected recommendations, but the benefits are smaller than that for the young populations.

Table 5(b) shows that unexpected recommendations would impose greater positive treatment effects for the premium users of Alibaba-Youku. This is the case because premium users have special access to the exclusive and high-quality video contents, thus more likely to receive attractive and unexpected video recommendations.

Table 5(c) shows that the gender difference of the treatment effects of unexpected recommendations is statistically insignificant. This suggest that there is no gender bias of user perception towards unexpectedness during the usage of the video platform.

Table 5(d) shows that users who live in big cities tend to value the unexpected recommendations less in comparison with those in the medium-level or small cities. This is probably due to the richer availability of entertainment in large cities, where people can easily get access to concerts or sports events, thus reducing their eagerness to explore the unexpected video content. On the other hand, the differences of treatment effects are not significant for those users living in the medium-level or small cities.

Table 5(e) shows that consumers using either the iOS or the Android operating system would enjoy better online experiences from the unexpected recommendation, compared to the consumers using other operating systems. In addition, we could witness even greater positive

treatment effects for those using the iOS operating system. The differences come from the heterogeneity of user demographics among different operating systems: 56.64% of the users using the iOS operating system are under 30 and more willing to explore, while only 38.47% of the users using the Android operating system are under 30; instead, 35.18% of them are above 60 and prefer to focus on their familiar content.

Overall, we find that these moderating factors can affect the treatment effects of our proposed model, shedding light on how to better design an unexpected recommender system. In particular, the heterogeneous treatment effects of unexpected recommendations on video consumptions with respect to user demographics are consistent across the three business metrics.

		CTR(1)	VV(2)	TS(3)
Treatment		0.0274*** (0.001)	0.0196*** (0.001)	0.7073*** (0.0937)
Panel (a)				
Treatment ×	Under 30	0.1146*** (0.002)	0.1149*** (0.002)	3.0990*** (0.2125)
	Between 30 and 60	0.0578*** (0.002)	0.0473*** (0.003)	1.3854** (0.3027)
	Above 60	-0.1384*** (0.003)	-0.1332*** (0.003)	-1.3855*** (0.3322)
Panel (b)				
Treatment × Premium		0.0124*** (0.002)	0.0250*** (0.002)	1.2375*** (0.1937)
Panel (c)				
Treatment × Female		-0.0089 (0.006)	-0.0091 (0.006)	-0.3274 (0.5889)
Panel (d)				
Treatment ×	Big City	-0.0915*** (0.002)	-0.0940*** (0.002)	-0.7890*** (0.2117)
	Medium City	-0.0006 (0.004)	-0.0005 (0.004)	0.4785 (0.3393)
	Small City	0.0011 (0.004)	-0.0038 (0.004)	-0.4886 (0.3593)
Panel (e)				
Treatment ×	Android	0.2447*** (0.009)	0.2264*** (0.009)	4.8660*** (0.5396)
	iOS	0.7600*** (0.009)	0.7584*** (0.009)	10.361*** (0.5700)
R-Squared		0.0006	0.0014	0.0006
Observations		26959538	26959538	26959538

*Table 5: Heterogenous Average Treatment Effect of the proposed model. The table shows a regression with robust standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01*

Discussions and Conclusions

Popularity of modern streaming services has stimulated the need to break the filter bubbles, where the users consume the old-and-tried types of videos, and provide them with fresh unexpected content lying outside of their comfort zones. In this paper, we present an unexpected recommendation method that models unexpectedness in the latent space and test its performance in an online controlled experiment conducted at Alibaba-Youku on over half a million customers for the period of one week. As a result, we demonstrate that unexpected recommendations significantly increase the click-through rate and the amount of video consumptions, and that the effect is the strongest when recommendations are not too unexpected, still taking the user outside of his or her comfort zone without pushing them into the realm of watching irrelevant videos. We have also demonstrated that unexpected recommendations are useful not for all the customers of Alibaba but only for certain segments of its population, such as young customers and premium users. This implies that unexpected recommendations should not be provided in the "shotgun" approach to everybody but only to the targeted segments of customers and only in a "measured" way without risking to make these unexpected recommendations irrelevant.

Furthermore, the improved business performance metrics in this experiment demonstrated strong economic effect of our method on Alibaba-Youku, leading the company to move our unexpected recommendation method into production.

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